

Kevin Pierce

PhD Student in Physical Geography

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Research Supervisor: Marwan Hassan

RECENT PUBLICATIONS

Pierce, J. K., & Hassan, M. A. (2020). Joint Stochastic Bedload Transport and Bed Elevation Model: Variance Regulation and Power Law Rests. *Journal of Geophysical Research: Earth Surface*, 125 (4), 1–15. doi: 10.1029/2019JF005259

Pierce, J. K., & Hassan, M. A. (2020). Back to Einstein: Burial-Induced Three-Range Diffusion in Fluvial Sediment Transport. *Geophysical Research Letters*, 47 (15), 1-10. doi: 10.1029/2020GL087440

Pierce, J. K., Ferreira, R. M. L., & Hassan, M. A. (2021). Langevin Approach to Bed Load Sediment Velocity Distributions. *Journal of Fluid Mechanics* (*Under Review*).

TEACHING ASSISTANT EXPERIENCE

PHYS 211: Introduction to Mathematical Physics (WVU – Dr. James Lewis – 192 hours) [Fall 2013]

PHYS 451: Introduction to Quantum Mechanics (WVU – Dr. James Lewis – 192 hours) [Spring 2013]

PHYS 101: Energy and Waves (UBC – Dr. Valerie Milner – 192 hours) [Fall 2015]

PHYS 110: Introductory Physics (UBC – Dr. Valerie Milner – 192 hours) [Spring 2016]

PHYS 304: Introductory Quantum Mechanics (UBC – Dr. Marcel Franz – 192 hours) [Fall 2016]

PHYS 403: Introductory Statistical Mechanics (UBC – Dr. Myra Tovar – 192 hours) [Spring 2017]

GEOB 103: Water and Landscapes (UBC – Dr. Brett Eaton – 192 hours) [Fall 2017]

GEOB 102: Climate and Ecosystems (UBC – Dr. Nina Hewitt – 192 hours) [Spring 2018]

GEOB 102: Climate and Ecosystems (UBC – Dr. Nina Hewitt – 192 hours) [Fall 2018]

GEOB 102: Climate and Ecosystems (UBC – Dr. Nina Hewitt – 192 hours) [Spring 2019]

GEOB 103: Water and Landscapes (UBC – Dr. Heather Chappells – 192 hours) [Summer 2019]

GEOB 103: Water and Landscapes (UBC – Dr. Matteo Saletti – 192 hours) [Fall 2019]

GEOB 103: Water and Landscapes (UBC – Dr. Matteo Saletti – 192 hours) [Spring 2020]

GEOB 103: Water and Landscapes (UBC – Dr. Heather Chappells – 192 hours) [Summer 2020]

GEOB 374: Statistics in Geography (UBC – Dr. Nina Hewitt – 192 hours) [Fall 2020]

GEOB 374: Statistics in Geography (UBC – Dr. Nina Hewitt – 192 hours) [Spring 2021]

GEOB 405: Fluvial Geomorphology (UBC – Dr. Marwan Hassan – 120 hours) [Spring 2020]

OTHER RELEVANT EXPERIENCE

Student Services Tutor: UBC Athletics [Fall 2015- Fall 2016]
Duties: Study hall style instruction and exam preparation for student athletes. Roughly 250 hours of 1-on-1 tutoring experience in math and physics (3 semesters, 28 three hour sessions per semester)

Student Services Tutor: WVU Access and Diversity [Fall 2012- Fall 2014]
Duties: 1-on-1 instruction and problem solving practice in mathematics (calculus), physics, and chemistry. Roughly 50 cumulative hours (2 one hour sessions per week for two semesters)

Private Calculus Tutor: Assorted UBC Students (references available) [Fall 2015-Spring 2016]
1-on-1 instruction in UBC calculus courses at all levels, including problem solving and concepts (Five students– 33 cumulative 1 hour sessions)

Online Course Development GA: UBC Geography [Summer 2020]
Developed interactive GEOB 103 laboratories for future semesters: Virtual field trip to Mt. Ranier in Tapestry in collaboration with Prof. Siohban McPhee. Video lecture preparation, Canvas content development, and construction of online lab assignments in Canvas (64 hour contract)

Online Course Development GA: UBC Geography [Spring 2021]
Integrated Github and Nbgrader technologies into a "Statistics in Geography" course to streamline online learning with Prof. Nina Hewitt. (92 hour contract).

Instructional Skills Workshop: UBC CTLT [Spring 2021]
Completed workshop on standard teaching models, active learning strategies, inclusive teaching, and portfolio development (Two week duration).

COMPLETED EDUCATION

- **Master of Science in Physics (Dr. Ian Affleck)** Vancouver, BC, Canada
University of British Columbia; Thesis: "Magnetic boundary structure of chiral graphene" [Fall 2014 – Spring 2016]
- **Bachelor of Science in Physics** Morgantown, WV, USA
West Virginia University; Thesis: "Physics of Random Phenomena" [Fall 2010 – Spring 2013]